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Photomorphogenic tomato mutants *high-pigment 1* and *aurea* responses to iron deficiency

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ABSTRACT

Iron (Fe) is a micronutrient for plant development, as constituent of several photosynthesis- and respiration-related proteins and enzymes. Consequently, Fe deficiency leads to chlorosis in leaves and plant growth impairment. It has become increasingly evident that light signals coordinate iron homeostasis in plants. To further address new insights into how light is a fundamental part of Fe deficiency responses, we employed Micro-Tom (wild type, WT) tomato (*Solanum lycopersicum* L.) plants and *high-pigment* 1 (*hp1*) and *aurea* (*au*) photomorphogenic mutants, which exhibit an excessive light response and low light perception, respectively. Plant growth, pigment contents, oxidative status, and nutrient profile were analyzed. The results revealed the influence of the different genotypes on Fe deficiency responses. WT and *au* exhibited plant growth reduction under Fe deficiency. WT, *hp1* and *au* demonstrated that Fe availability and light perception play fundamental roles in chlorophyll and anthocyanin biosynthesis. Lipid peroxidation was not increased for any genotype under Fe deficiency, indicating that mutations in light perception and signaling differentially modulate H_2O_2 production and scavenging under this condition. Additionally, macronutrients and micronutrients were taken up and distributed differently among the different plant genotypes, tissues and Fe conditions analyzed. In general, the *au* plants accumulated lower amounts of nutrients (Ca, S, P, Mg, B and Zn) than the WT and *hp1* genotypes regardless of the Fe concentrations. Our data clearly indicates that light perception and signaling influence Fe-dependent morphophysiological responses in plants, suggesting possibilities for biotechnological improvement of crops grown under Fe shortage.

1. Introduction

Iron (Fe) is a plant essential micronutrient, participating in fundamental metabolic processes, especially due to its ability to change its redox state (Nikolic and Pavlovic 2018). Fe takes part in important metalloproteins or it is required as cofactor for redox enzymes in the photosynthetic and respiratory electron transport chain in the chloroplasts and mitochondria (Connorton et al., 2017). Fe is also essential for

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chlorophyll *b*iosynthesis and stabilization, activity of antioxidant enzymes, and biological nitrogen fixation. (Zuchi et al., 2015; López-Millán et al., 2016; Lucena and Hernandez-Apaolaza 2017; Hura et al., 2018).

Despite widely abundant in the soil, Fe solubility and availability for plants is affected by soil pH and aeration, redox potential, microbial processes, and presence of other minerals, being often insufficient, especially in calcareous and alkaline soils (Vose, 1982; Zhang et al., 2019). The low availability makes Fe deficiency an world-wide yield-limiting factor for many crops (Zuo and Zhang 2011). Fe was shown to be a limiting factor for biomass and seed yield in Arabidopsis (*Arabidopsis thaliana*) (Ravet et al., 2009) and in several crops including tomato (*Solanum lycopersicum*) (Jin et al., 2009), spinach (*Spinacia oleracea*) (Jin et al., 2013), rice (*Oryza sativa*) (Bashir et al., 2011) and soybean (*Glycine* max) (Merry et al., 2022). Early symptoms of Fe shortage include interveinal chlorosis of young leaves, and consequential reduced CO_2 fixation and respiration leads to plant growth impairment, especially in the roots (Fu et al., 2017; Satbhai et al., 2017). Fe deficiency is known to alter photosynthetic rate, chloroplast structure and chlorophyll synthesis, the latter explaining the interveinal yellowing of leaves (Briat et al., 2015). As cofactor of important antioxidant enzymes, like catalase and ascorbate peroxidase, Fe is directly related to the plant oxidative status under stressful conditions (Tripathi et al., 2018). Several studies point out the ameliorative effect of Fe on reducing the damages provoked by reactive oxygen species (ROS) under high salinity (Ghasemia et al., 2014) and drought conditions (Akbari et al., 2013).

Iron in the rhizosphere is mainly present as ferric (Fe^{3+}) low-solubility oxyhydrates, not directly accessible to plants. Different species had, therefore, evolved specialized strategies for iron uptake (Römheld and Marschner, 1986; Connorton et al., 2017). Non-gramineous plants, such as tomato and *Arabidopsis* have served as models for reducing strategy (Strategy I), in which plasma-membrane *H*+/ATPases release protons in the rhizosphere, acidifying the surrounding environment, and triggering the reduction of Fe^{3+} by FERRIC REDUCTION OXIDASE 2 (FRO2) enzymes at the plasma membrane to the ferrous ionic form (Fe^{2+}) absorbable by root cell transporters such as IRON-REGULATED TRANSPORTER (IRT1) (Robinson et al., 1999; Vert et al., 2002). Gramineous plants, such as maize and rice, employ a chelating strategy (Strategy II), in which Fe^{3+}-high affinity phytosiderophores are secreted in the rhizosphere. The chelated complex is then recognized and taken up by specific transporters into the root cells (Takagi et al., 1984; Zhang et al., 2019).

The extent of the abovementioned strategies is modulated concerning Fe availability in soils to maintain iron homeostasis, defined as the balance between uptaking sufficient Fe amounts to meet metabolic needs without accumulating toxic levels in the tissues (Connorton et al., 2017). Under Fe deficiency, plants maximize Fe uptake from the soil by modifying root architecture with increased number of lateral roots and density of hair cells (Li et al., 2016), and up-regulated expression of genes encoding root H^+/ATPases, FRO2 and IRT1 (Yuan et al., 2008; Zhang et al., 2019). There is also an interesting crosstalk between Fe and other mineral nutrients, especially under Fe-deficient conditions (Nikolic and Pavlovic, 2018). Low Fe availability induces the expression of genes related to the transport and assimilation of phosphorus (P) and sulfur (S) (Briat et al., 2015), and increased the concentrations of other divalent cations, such as zinc (Zn), copper (Cu), manganese (Mn) and cobalt (Co) (Gayomba et al., 2015), suggesting that the whole plant metallome is altered when Fe is in shortage.

Unraveling Fe-sensing status systemic signals is a longstanding challenge, since many metabolic and genetic mechanisms of Fe-dependent responses remain elusive. It is well known, however, that Fe homeostasis in plant organs is affected by environmental conditions, such as light (Hu et al., 2013; Vélez-Bermúdez and Schmidt 2022). Fe-sensing and Fe-dependent adjustments of the plant circadian clock are coordinated by the red-light photoreceptors phytochromes (PHYs) (Salomé et al., 2013). Under Fe deprivation conditions, the PHYB-dependent accumulation of the phloem-mobile photoresponsive transcription factor ELONGATED HYPOCOTYL 5 (HY5) was suggested to be a mobile signal from the Fe-sink leaves to induce the expression of important Fe uptake genes in roots (Guo et al., 2021; Vélez-Bermúdez and Schmidt 2022). These findings prompt new questions regarding how morphophysiological responses to Fe are directly modulated by light signaling.

Thus, we hypothesized that the sensibility and activation of light signaling-dependent pathways are correlated to the magnitude of Fe deficiency morphophysiological responses. For this purpose, we investigated the effects of Fe deficiency over growth, oxidative status, and nutrient profile on two contrasting photomorphogenic tomato mutants: *high-pigment* 1 (*hp1*), which show excessive responses to events mediated by light (Kendrick et al., 1997), and *aurea* (*au*), functional PHY-deficient (Terry and Kendrick, 1996; Muramoto et al., 2005). By characterizing the impacts of light signaling mutations on Fe deficiency responses, we may shed light over possibilities of crop manipulation to cope with such yield-limiting factor.

2. Material and methods

2.1. Plant material, growth conditions and iron deficiency treatment

Seeds of the photomorphogenic tomato mutants (*Solanum lycopersicum* L.) high-pigment 1 (*hp1*), aurea (*au*), and their wild-type background, Micro-Tom (WT), were germinated in a growth room under controlled temperature (25 ± 2 °C) and relative humidity (50% ± 10%), with 12-hour photoperiod providing 250 μmol photons m^{-2} s^{-2} (Philips® fluorescent lamps 58 W). Germination trays contained a 1:1 mixture of commercial substrate (Plantmax HT, Eucatex, Brazil) and expanded vermiculite, supplemented with 1 g L^{-1} 10:10:10 NPK and 4 g L^{-1} lime ($MgCO_3 + CaCO_3$ pa). Ten days after germination, under the same conditions previously described, seedlings were transferred to polypropylene pots with 180 mL capacity filled with Hoagland and Arnon nutrient solution (NS) with 50% of ionic strength, continuously aerated. Each pot had five plants, floating on the NS fixed in holes of expanded polystyrene (Isopor®) plates by polyurethane foam strips that were placed around the plant collar. After five days acclimatization period, control plants were transferred to complete Hoagland and Arnon NS, while treatment plants were exposed to NS withdrawn Fe, and the pots were randomly arranged on the shelves. The complete nutrient solution contained 1.0 mM KH_2PO_4, 2.5 mM KNO_3, 2.0 mM $Ca(NO_3)_2$, 0.75 mM $MgSO_4$, 10 μM H_3BO_3, 1.0 μM $MnCl_2$, 0.5 μM $ZnSO_4$, 0.5 μM $CuSO_4$, 0.05 μM H_2MoO_4 and 20 μM $C_{18}H_{16}O_6N_2NaFe$ (Fe-EDDHA). The NS pH was adjusted to 5.8 daily and the whole NS was changed every 3 days. All the measurements were proceeded at harvesting, after 10 days of Fe deficiency.

2.2. Plant growth measurements

Root and shoot length were measured using a digital caliper. Visual symptoms of deficiency were photographed and described. To calculate the branching index, the ratio of the sum of the length of each lateral branch and the central axis was considered, according to Morris et al. (2001). The total leaf area per plant was measured with an area meter (LI-3100C, LI-COR, USA). The roots and shoots were dried for 48 h at 50 °C, and then weighed on an analytical scale for quantification of biomass accumulation.

2.3. Nutrient profile of leaves and roots

After weighing, the dry biomass of the shoots and roots were used to determine the macro and micronutrient contents. 200 mg of dry weight was microwave-digested with 2 mL 70% HNO_3, 2 mL H_2O_2 and 2 mL Milli-Q water (18.2 MΩ cm a 25 °C) at a controlled pressure of 2 MPa

(Chilimba et al., 2011). The digested samples were diluted to 10 mL with Milli-Q water and analyzed by ICP-OES chromatograph (Optima 8300, Perkin Elmer, USA).

2.4. Pigment quantification

Four leaf discs (25 mg) were used for total chlorophyll (Chl) determination as described by Lichtenthaler (1987). The extraction was

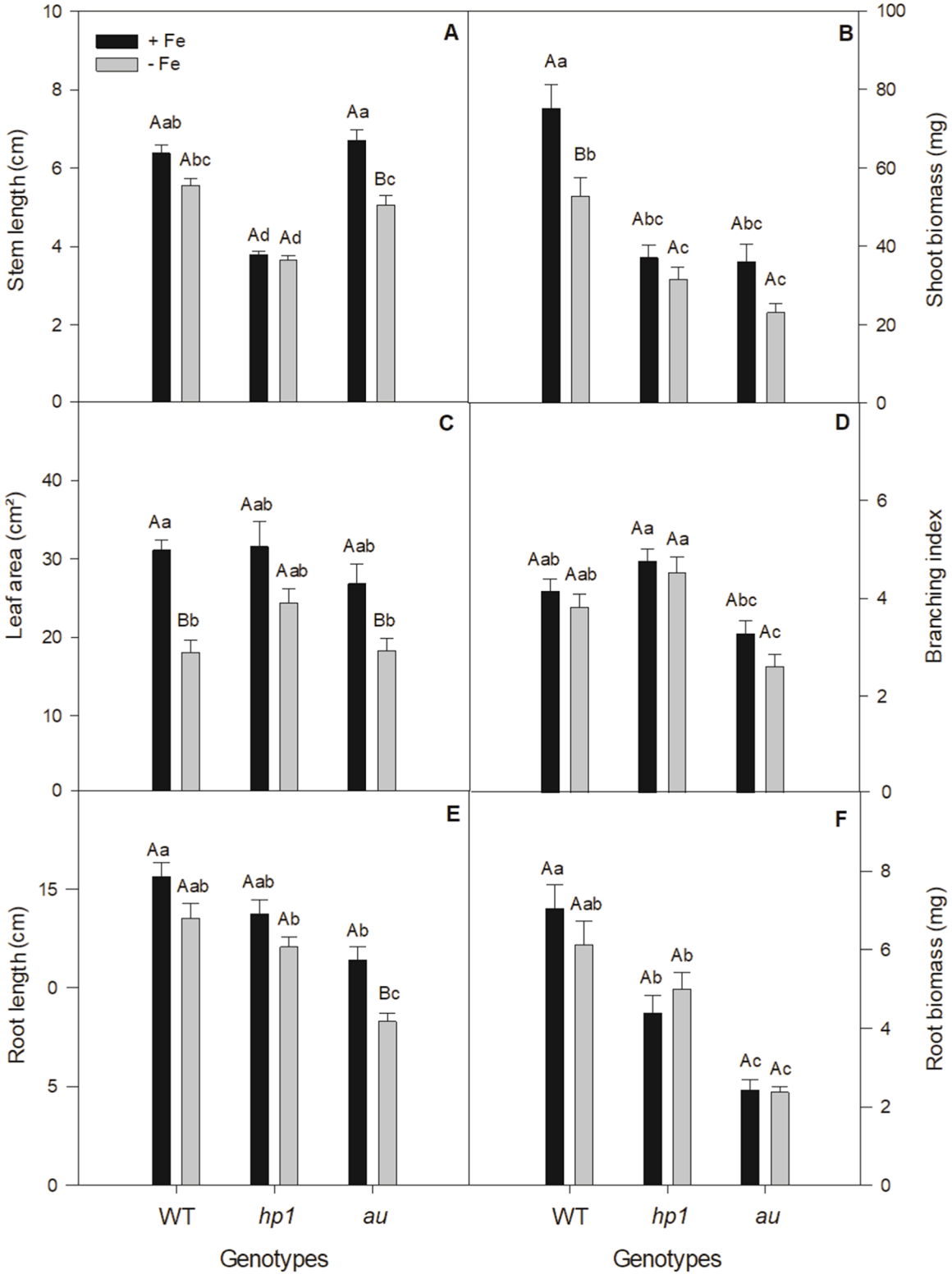


Fig. 1. Morphological responses of wild-type tomato cv. Micro-Tom (WT) and photomorphogenic mutants *high-pigment* 1 (*hp1*) and *aurea* (*au*) to Fe deficiency. (A) Stem length, (B) shoot biomass, (C) leaf area, (D) branching index, (E) root length, (F) root biomass. Bars are mean ± SEM ($n = 4$). Distinct uppercase letters indicate significant differences between Fe treatments while distinct lowercase letters indicate significant differences between genotypes (Two-way ANOVA followed by Tukey HSD test, $\alpha = 0.05$).

carried out by adding 1.5 mL of 100% acetone. Anthocyanins were extracted using 25 mg of hypocotyls fresh mass diluted in a solution of hydrogen chloride:water:chloroform (24:18:48, v/v/v). Tubes were kept at 4 °C in the dark for 24 h. The hypocotyls were then removed, and the remaining extract was evaluated according to Peters et al. (1992). All extraction processes were performed in triplicate. The absorbance of the extracts was measured in a spectrophotometer (DU 640, Bekman, USA) at wavelengths of 645 nm (Chl b), 663 nm (Chl a), 470 nm (carotenoids) and 535 nm (anthocyanin).

2.5. Lipid peroxidation and H_2O_2 content measurements

Lipid peroxidation was determined by estimating the content of thiobarbituric acid reactive substances (TBARS) in 100 mg of plant tissue, following the method of Heath and Packer (1968) adapted by Alves et al. (2016). The concentration of malondialdehyde (MDA) was calculated using a coefficient of absorbance of 1.55×10^{-5} mol^{-1} cm^{-1}, with readings between 535 and 600 nm. The H_2O_2 content was determined as described by Alexieva et al. (2001). 100 mg of fresh mass were homogenized in 0.1% (m/v) trichloroacetic acid (TCA), the homogenate was centrifuged at 12.100 x g at 4 °C for 15 min, and 200 μL of the supernatant was added to 200 μL of 100 mM K-phosphate buffer (pH 7.0) plus 800 μL of 1 M KI. The absorbance was read at 390 nm.

2.6. Data analyses

A completely randomized design was performed and for each treatment we used seven replicates (five plants each). Results were expressed as the mean and standard error of the mean (±SEM). Growth parameters, branching index and nutritional quantification were measured using four replicates (20 plants). Pigments, lipid peroxidation (MDA) and H_2O_2 quantification were assessed using three replicates (15 plants). Means were compared by two-way analysis of variance (ANOVA) between three genotypes (TR1) and Fe treatments (TR2), using Tukey's honestly significant difference (HSD) test ($\alpha = 0.05$. Statistical analyses were performed in the R software (R Core Team, 2020). P-values of homogeneous groups (TR1 or TR2) and all-pairwise comparisons of TR1 and TR2 interactions were presented in Table S1.

3. Results

3.1. Fe deficiency impacts over plant growth

Fe deficiency differentially affects the morphological development of the photomorphogenic tomato mutants *au* and *hp1*, despite their usual discrepancies due to genotype effects, compared to WT when grown under no Fe limitation.

hp1 displays the shortest stems between the genotypes under control conditions, however, no further reduction is registered when cultivated under Fe limitations, in opposition to the lower stem length displayed by *au* plants (Fig. 1A). As for aerial biomass, Fe deficiency reduced shoot weight in tomato WT, but no similar effect was noted in the already lighter shoots of the photomorphogenic mutants (Fig. 1B). Fe deprivation significantly reduced WT leaf area (Fig. 1C) and reinforced the difference concerning the lower *au* branching index compared to the other genotypes (Fig. 1D). The visual aspects of the impacts of Fe

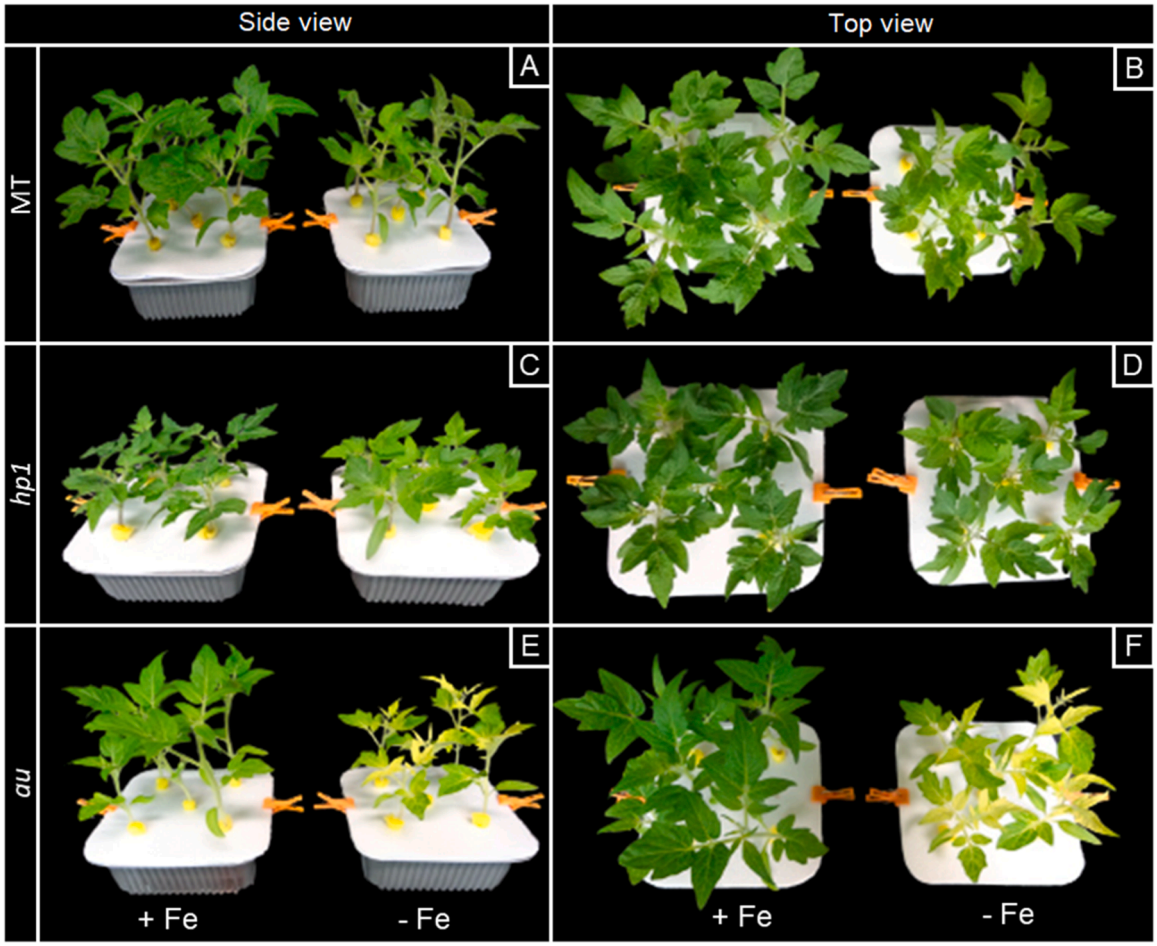


Fig. 2. Side and top views of shoot morphology aspects of wild-type tomato cv. Micro-Tom (WT, A-B) and photomorphogenic mutants *high-pigment* 1 (*hp1*, C-D) and *aurea* (*au*, E-F) in presence (+) and absence (-) of Fe.

deficiency over shoot morphology can be consulted on Fig. 2 for the tomato genotypes.

Root length reduction was observed in *au* plants, a condition further intensified by Fe deficiency (Fig. 1E). On the other hand, Fe starvation did not affect directly root biomass in any genotype (Fig. 1F).

Taken together, these results suggest that *hp1* morphological development seems not to be affected by Fe limitation, whereas WT impairs leaf area growth and stem biomass accumulation and *au* inhibits root and shoot elongation when Fe is deprived.

3.2. Fe deficiency effects on pigment accumulation and antioxidant metabolism

Under adequate Fe supply, chlorophyll a and b content were similar between WT and *hp1* leaves, while *au* presented the lowest levels. No differences were registered for carotenoid content between the genotypes. As for anthocyanins, highest levels were found in *hp1* leaves and the lowest in *au* plants (Fig. 3).

Fe deficiency reduced pigment levels in tomato plants, however, the effects were distinctive for each genotype. *au* leaves presented lower content of chlorophyll a, and WT leaves reduced levels of both chlorophyll a and b when Fe was deprived. No effect on chlorophyll levels was registered for *hp1*, however, it was the only genotype to show decreased anthocyanin content due to Fe deficiency. All genotypes presented lower accumulation of carotenoid content in the leaves, being extremely reduced in *au* plants (Fig. 3).

Alterations of H_2O_2 levels due to Fe deficiency in tomato plants depends on the organ and the genotype. Under adequate Fe supply, no differences in the H_2O_2 content were registered between the genotypes' leaves, while *au* roots presented lower levels compared to WT and *hp1* roots. When Fe is deprived, H_2O_2 content is increased in WT and *hp1* leaves, while in the roots, its levels are reduced for all genotypes, being lower for the photomorphogenic mutants compared to WT under the same conditions (Fig. 4A, B).

Despite the effects of Fe deficiency over H_2O_2 content, no correlated effects were registered concerning MDA levels, neither in leaves nor roots for any genotype (Fig. 4C, D).

3.3. Nutrient profile is altered in the au and hp1 photomorphogenic mutants

Macronutrients and micronutrients profiles were assessed in the leaves and roots of all genotypes (Table 1). Under adequate Fe supply, WT and *hp1* leaves presented a similar nutrient profile. Compared to WT, *au* leaves presented lower levels of Fe, P, Ca, Mg and B, in spite of accumulating more K. Concerning the roots, lower contents of Fe, P, Mg,

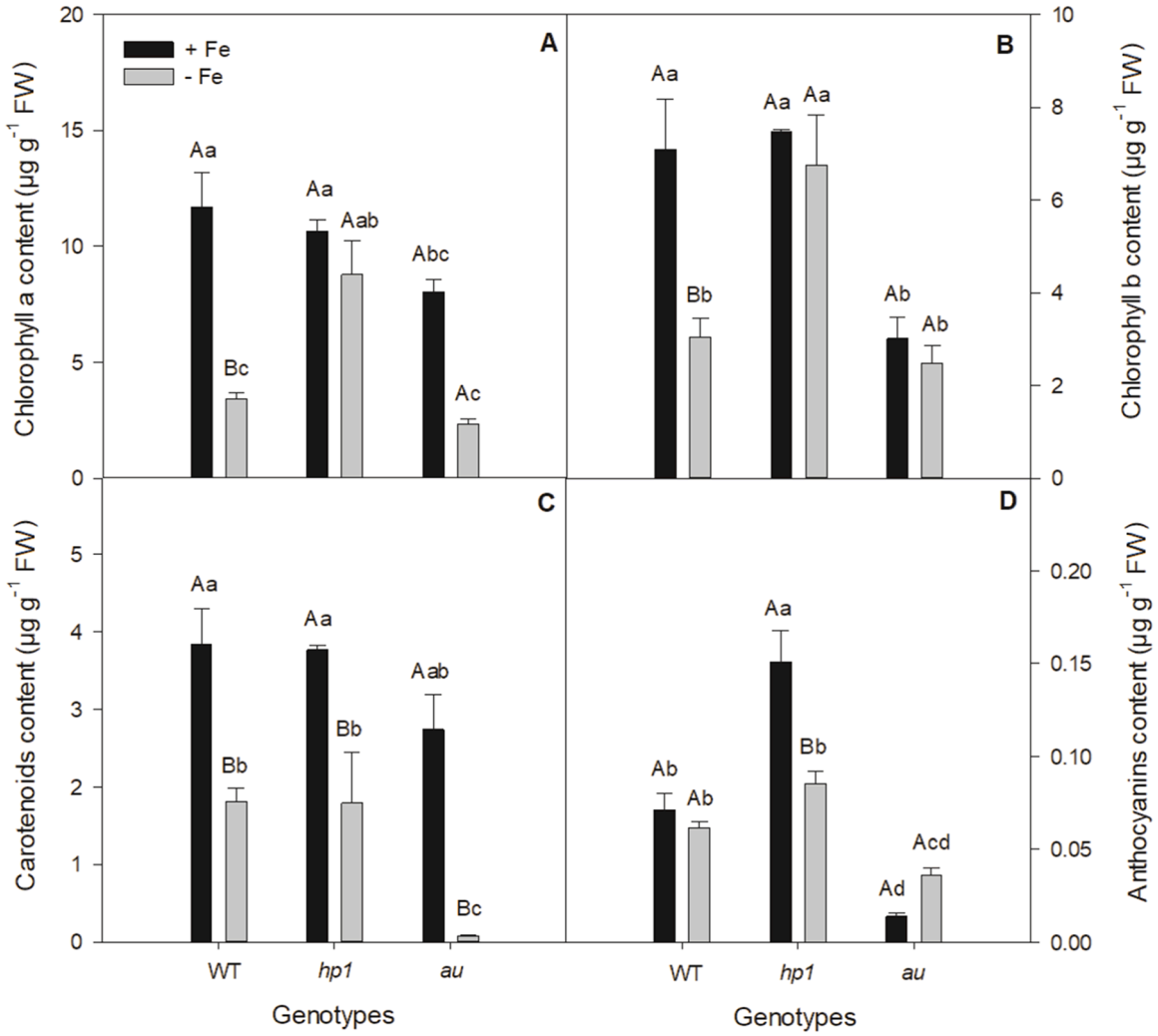


Fig. 3. Pigment content of wild-type tomato cv. Micro-Tom (WT) and photomorphogenic mutants *high-pigment* 1 (*hp1*) and *aurea* (*au*) grown in presence (+) and absence (-) of Fe. (A) Chlorophyll a, (B) chlorophyll *b*, (C) carotenoids, (D) anthocyanins. Bars are mean ± SEM ($n = 3$). Distinct uppercase letters indicate significant differences between Fe treatments while distinct lowercase letters indicate significant differences between genotypes (Two-way ANOVA followed by Tukey HSD test, $\alpha = 0.05$).

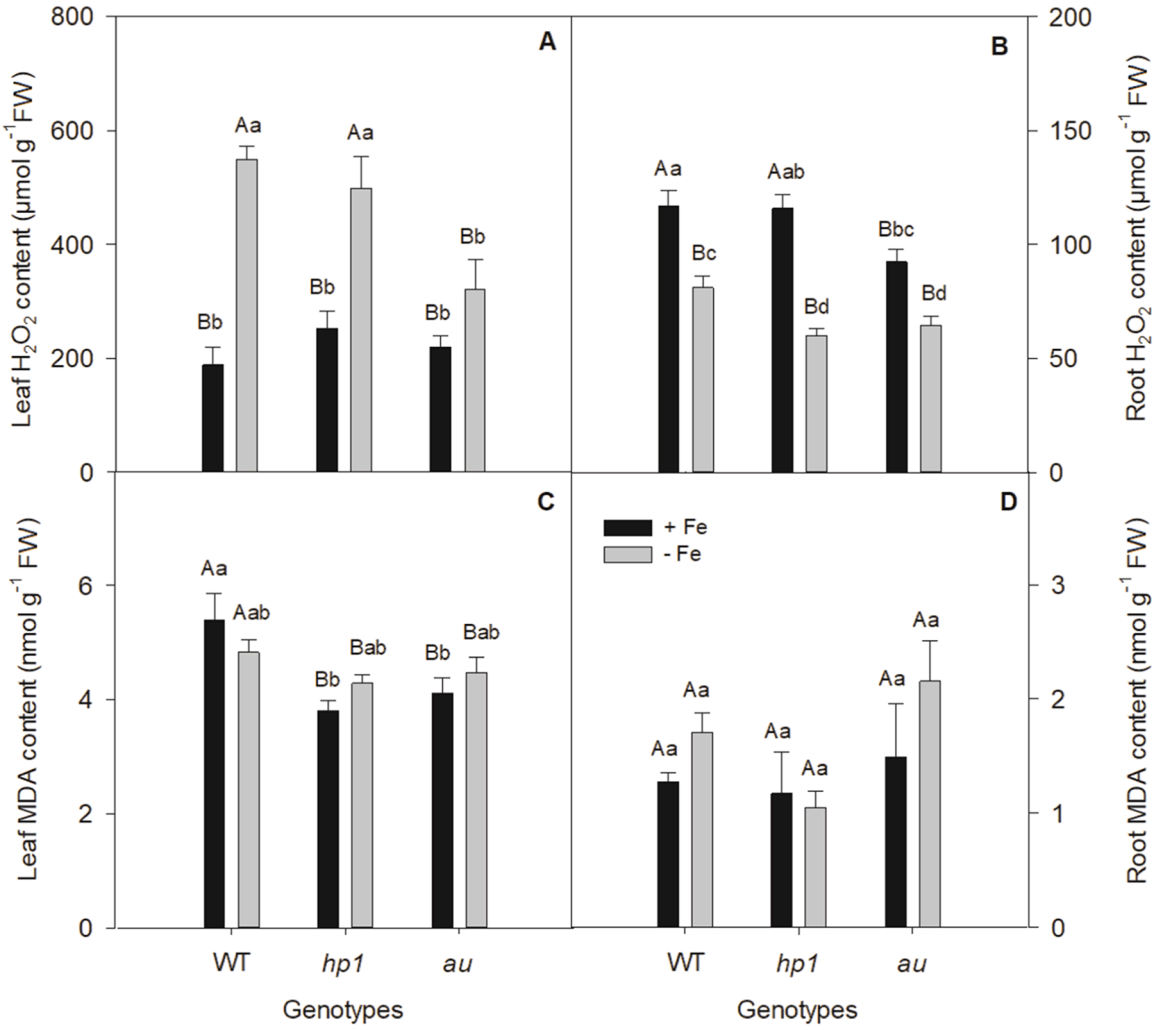


Fig. 4. Oxidative status of wild-type tomato cv. Micro-Tom (WT) and photomorphogenic mutants *high-pigment* 1 (*hp1*) and *aurea* (*au*) grown in presence (+) and absence (-) of Fe. Hydrogen peroxide (H_2O_2) content in leaves (A) and roots (B), malondialdehyde (MDA) content in leaves (C) and roots (D). Bars are mean ± SEM ($n = 3$). Distinct uppercase letters indicate significant differences between Fe treatments while distinct lowercase letters indicate significant differences between genotypes (Two-way ANOVA followed by Tukey HSD test, $\alpha = 0.05$).

Mn and Zn were registered for *au*, and lower contents of Ca and B were determined for both photomorphogenic mutants.

When grown under Fe deficiency, lesser content of Fe was registered in the leaves of all genotypes. The only modifications in the nutrient profile reported in the leaves for each genotype due to Fe limitation were the accumulation of Mn and Zn in WT and *hp1*, and the shortage of B in *au* plants.

WT and *au* roots presented reductions in Fe content due to Fe deficiency, and interestingly, Fe levels in the *hp1* roots were similar in both treatments. Fe deficiency increased Cu content in all genotypes, and B content in *au*. Fe limitation led to lower content of P and B in WT, and Ca in both WT and *hp1*.

The results suggest that despite the differences concerning the nutrient profile of the photomorphogenic mutants, the light signaling mutations also interfere in the accumulation of other nutrients when Fe is limited.

4. Discussion

Tomato photomorphogenic mutants have been extensively used to unravel the mechanisms by which light signaling control plant development (Carvalho et al., 2011). The aurea (*au*) genotype is a phytochromobilin synthase mutant (Muramoto et al., 2005), being deficient in functional phytochromes, especially during early weeks of plant development (Terry and Kendrick 1999). On the other hand, the high-pigment 1 (*hp1*) genotype presents a mutation in the *UV-DAMAGED DNA BINDING PROTEIN 1* (*DDB1*), an early repressor of light signal transduction pathway (Lieberman et al., 2004), characterized by excessive photoresponsiveness. Briefly, active PHYs represses the activity of DDB1-constituent ubiquitin ligase complex, which degrades transcription factors (such as HY5) related to photodependent responses as chlorophyll *b*iosynthesis, de-etiolation, and nutrient uptake (Legris et al., 2019; Xiao et al., 2022). Accordingly, *au* presents yellow-green leaves, lower content of chlorophyll and anthocyanins, and elongated hypocotyls, while *hp1* leaves are dark green due to accumulation of photosynthetic pigments and its hypocotyls are relatively short (Kendrick et al., 1997). These two contrasting mutants were used in this present work to better investigate the involvement of light signaling on the Fe deficiency responses.

Taking part of diverse regulatory proteins involved in chlorophyll synthesis and stabilization, electron carriers during photosynthesis and respiration (Hänsch and Mendel, 2009; Devadasu et al., 2016; López-Millán et al., 2016), it is not surprising that under Fe deficiency conditions, plant growth parameters are severely impaired (Briat et al., 2015; Rout and Sahoo 2015; López-Millán et al., 2016). We observed lower chlorophyll levels due to Fe deprivation were associated with

Table 1
Nutrient profile of leaves and roots of wild-type tomato cv. Micro-Tom (WT) and photomorphogenic mutants *high-pigment* 1 (*hp1*) and *aurea* (*au*) grown in presence (+) and absence (-) of Fe. Data are mean ± SEM ($n = 4$). Statistical difference between means are given by different letters for each organ (Two-way ANOVA followed by Tukey HSD test, $\alpha = 0.05$).

Treatments	P mg g^{-1}	K	Ca	Mg	S	Fe µg g^{-1}	Cu	Mn	Zn	B
	Leaf									
MT +Fe	11.9 ± 0.2a	60.4 ± 2.9c	39.7 ± 4.7a	11.6 ± 0.6a	10.2 ± 0.8a	85.6 ± 3.7a	16.8 ± 0.5ab	276.4 ± 31.0bc	37.4 ± 4.3b	210.1 ± 7.5a
MT -Fe	12.6 ± 1.6a	65.7 ± 1.6bc	40.3 ± 2.1a	11.7 ± 0.1a	9.3 ± 0.1ab	60.8 ± 1.6b	22.8 ± 1.0a	480.3 ± 32.1a	129.3 ± 13.4a	187.4 ± 6.0a
hp1 +Fe	9.1 ± 0.8ab	62.1 ± 0.9c	32.2 ± 2.8ab	9.9 ± 1.0ab	8.0 ± 0.6ab	75.5 ± 4.1ab	15.4 ± 1.0ab	186.2 ± 15.8c	73.6 ± 4.4b	179.9 ± 11.0a
hp1 -Fe	8.9 ± 0.7ab	56.2 ± 4.0c	31.9 ± 2.5ab	9.4 ± 0.1ab	8.0 ± 0.1ab	42.1 ± 4.0c	20.0 ± 0.2ab	416.2 ± 38.7ab	128.1 ± 23.8a	152.3 ± 2.1ab
au +Fe	8.1 ± 1.6b	81.5 ± 3.3a	24.9 ± 2.7b	5.9 ± 1.2c	6.0 ± 1.4ab	57.8 ± 2.5bc	11.0 ± 0.9b	166.3 ± 18.7c	55.2 ± 1.4b	144.4 ± 10.3b
au -Fe	6.4 ± 0.3b	75.7 ± 3.6a	17.1 ± 0.3b	3.5 ± 0.1c	4.4 ± 0.1b	20.0 ± 1.41d	11.6 ± 0.9b	228.2 ± 47.5bc	70.0 ± 34.2b	109.8 ± 5.2c
	Root									
MT +Fe	6.5 ± 0.3a	57.7 ± 2.4ab	13.9 ± 0.6a	1.4 ± 0.1a	0.7 ± 0.7ab	148.1 ± 1.6ab	5.9 ± 0.7b	449.1 ± 24.5a	258.0 ± 5.7a	28.2 ± 5.2a
MT -Fe	1.3 ± 0.1b	67.8 ± 1.6a	1.0 ± 0.1c	1.2 ± 0.1ab	0.9 ± 0.1a	85.0 ± 2.5c	27.5 ± 1.7a	307.1 ± 23.1ab	295.1 ± 15.0a	11.4 ± 1.5bc
hp1 +Fe	2.9 ± 1.1ab	57.4 ± 2.2ab	5.8 ± 4.7b	1.0 ± 1.3ab	0.6 ± 0.6ab	163.3 ± 8.5a	3.6 ± 3.1b	302.5 ± 57.4ab	232.2 ± 17.7ab	14.8 ± 15.8bc
hp1 -Fe	1.0 ± 0.1b	51.4 ± 2.5b	0.7 ± 0.1c	1.6 ± 0.3a	0.9 ± 0.1a	163.4 ± 0.5a	26.5 ± 1.3a	267.5 ± 26.4b	249.8 ± 31.9a	9.6 ± 0.5bc
au +Fe	0.5 ± 1.2b	52.4 ± 0.5b	1.0 ± 2.4c	0.3 ± 0.3c	0.3 ± 1.1b	65.5 ± 1.4c	3.2 ± 0.8b	106.7 ± 17.4c	82.8 ± 5.3c	4.5 ± 1.0c
au -Fe	0.6 ± 0.1b	51.1 ± 0.6b	0.5 ± 0.1c	0.5 ± 0.1bc	0.4 ± 0.1ab	46.5 ± 1.4d	28.6 ± 0.5a	166.0 ± 12.8c	124.4 ± 6.2bc	18.1 ± 1.2ab

altered growth patterns in WT and *au* tomato plants. WT reduced leaf area and shoot biomass while *au* shortened stem and root lengths. *hp1* chlorophyll levels, on the other hand, were not affected by Fe deficiency, and correspondingly, no negative effects on plant growth and morphology were registered. It seems that the greater chloroplast density and chlorophyll accumulation caused by *DDB1* mutation (Cookson et al., 2003; Caspi et al., 2008; Alves et al., 2016) could overcome the detrimental effects of Fe deprivation.

Required as cofactors of heme-dependent reactive oxygen species (ROS) scavenging enzymes, the deficiency of Fe can lead to accumulation of H_2O_2 inside plant cells (Galaris et al., 2019; Santos et al., 2019). Fe deprivation led to increased H_2O_2 content in WT and *hp1* leaves but not in *au*. In the roots under Fe deficiency, H_2O_2 content were lower for both photomorphogenic mutants compared to WT. It may be explained due to the higher tolerance presented by *hp1* and *au* to cope with oxidative stress. *au* plants presents higher activity of H_2O_2-scavenging peroxidases while *hp1* seems to rely upon non-enzymatic antioxidant compounds to remove ROS (Alves et al., 2016). It is interesting to notice an organ-dependent effect of Fe deficiency leading to H_2O_2 accumulation in leaves but reduction in the roots. MDA levels, an indicator of lipid peroxidation due to action of ROS over cell membranes, were not affected by the respective alterations of H_2O_2 levels, suggesting that Fe deprivation treatment was not severe enough to inflict significant cell damage. When Fe is scarce, there is an increased participation of other divalent cations, such as Cu, Zn, and Mn, as prosthetic metals in the active sites of ROS-scavenging enzymes (Waters and Armbrust 2013; Puig 2014). In line with this, the accumulation of Zn and Mn in the leaves of WT and *hp1*, and Cu in the roots of all genotypes in substitution of Fe in the composition of the antioxidant enzymes might have helped to contend ROS burst and consequent increased MDA levels.

Lower levels of nutrient content were registered in the organs of *au* even when Fe was not limited. Since *au* is not impaired in photosynthetic efficiency (López-Juez et al., 1990; Alves et al., 2016), which could impair the adequate supply of ATP and carbohydrates for the very energy-consuming nutrient absorption and transport processes, it highlights the strong and direct influence of phytochrome-mediated light signaling over the regulation of many nutrient uptake-related enzymes and transporters (Sakuraba and Yanagisawa 2018; Xu et al., 2021; Soares et al., 2021). P uptake was severely reduced in phytochrome-mutant rice seedlings (Sakuraba and Yanagisawa 2020), and, in *Arabidopsis*, P acquisition was promoted by stabilization of HY5, which, in turn, increased the expression of important P transporters (Sakuraba et al., 2018). As for Fe, Guo et al. (2021), indicated that under low supply, there is a PHYB-dependent HY5-mediated induction of important Fe transporters and, accordingly, PHY mutants presented lower content of Fe in their tissues. Since some Fe transporters can also uptake other divalent metal cations (Grotz and Guerinot 2006; Krishna et al., 2020; Rai et al., 2021), the lower levels of Ca, Mg, Mn, and Zn in *au* roots may be due to the absence of PHY-dependent induction of such transporters. Correspondingly, the contents of the mobile nutrients Mn and Zn were not increased in *au* leaves under Fe deficiency, in opposition to WT and *hp1*.

hp1 was able to maintain similar Fe content in the roots regardless of the availability conditions, indicating that light signaling may positively affect iron homeostasis and maintenance under deprivation conditions beyond the aforementioned PHY-dependent induction of Fe transporters (Guo et al., 2021). Increased levels of nutrient absorption were not registered by *hp1* compared to the WT, contrasting with the severe negative effects of the *au* mutation over nutrient profile. It may occur due to the distinct nature of the mutations. *au* impact over phytochromobilins affects the activity of all phytochromes, tending to globally hinder PHY-dependent responses. *hp1*, on the other hand, is a mutation of a single repressor acting downstream the phytochromes in the light signaling pathway. Moreover, other photomorphogenic repressors are still active, restraining the response magnitude, especially with those related to the increased HY5 activity under Fe deficiency.

5. Conclusions

Our findings corroborate the growing evidences of light- and PHY-influences over nutrient uptake and homeostasis in plants. PHY-deficient *au* tomato plants were more sensitive to iron deficiency, presenting shorter stems and roots, and they were also unable to increase the content of Mn and Zn in the leaves. This points out that further investigations could be focused on PHY signaling biotechnological manipulation to develop crop cultivars that can cope with iron

deficiency in the soils. Regarding *hp1* mutant, the loss of function of the photomorphogenic repressor DDB1 in this genotype allowed plants to overcome chlorophyll deficiency due to Fe shortage, with no constraints to their normal morphological development. Light exaggerated response in tomato genotypes has revealed many important agronomic traits, such as enhanced antioxidant activity compounds (Pal et al., 2019), featuring compelling potential towards further studies aimed to genetically explore this genotype, especially concerning productivity in Fe-deficient soils.

Authorship contribution statement

RF Carvalho & CC Monteiro contributed to the study conception and design. Material preparation, data collection and development analysis were performed by CC Monteiro MA Gavassi & LA Gaion. Antioxidant analyses were performed by LR Alves & PL Gratão. Nutritional analysis was performed by RM Prado. The first draft of the manuscript was written by MA Gavassi & FRR Alves and all authors have written, critically revised, and discussed all aspects in the manuscript. All authors read and approved the final manuscript.

CRediT authorship contribution statement

Marina Alves Gavassi: Writing – original draft. **Frederico Rocha Rodrigues Alves:** Writing – original draft. **Carolina Cristina Monteiro:** Methodology. **Lucas Aparecido Gaion:** Methodology. **Letícia Rodrigues Alves:** Methodology. **Renato de Mello Prado:** Writing – review & editing. **Priscila Lupino Gratão:** Writing – review & editing. **Rogério Falleiros Carvalho:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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